

Kindled — Encounter Design Crew

Six agents that turn measured project canon into a DataTable-ready encounter specification, priced against a hard frame budget.

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Crew: crew/kindled_crew.py 28 July 2026

Where each criterion is answered

CRITERION	EVIDENCE IN THIS SUBMISSION	PTS
Working Crew	Six agents run end to end with no crash. Latest run emitted 7 DataTable rows after 1 negotiation round — §5, with the full agent transcript in crew/out/run-report.md.	3.0
Game Connection	Built for Kindled , the capstone. Output is consumed by the UE project in this same repository; the crew reads that project's own shipped design and <i>measured</i> performance files — §1 and §6.	3.0
Role Clarity	Per-agent role, inputs and outputs — §4 — plus what breaks if each is removed.	2.0
Architecture Diagram	Mermaid flowchart TD in crew/README.md, rendered at §3.	1.0
ReadMe	crew/README.md — names the game, states the crew's purpose, describes what it produces.	1.0

1 · The game, and why encounters are a budget problem

Kindled is a top-down roguelike in Unreal Engine 5.8. You carry the only fire in a pitch-dark world, and an army gathers in your light because outside it they die. You command that army through four broad stances rather than unit selection — three to ten units early, hundreds by late run.

That premise makes encounter design a **budget question, not a taste question**. Every enemy on screen costs draw time, and the project has a measured cost curve for exactly that (docs/perf/BUDGETS.md). The ceiling is hard: **16.6 ms per frame**. A wave that reads well but cannot be drawn is not a design — which is why this crew contains an agent whose only power is to reject the designer's work.

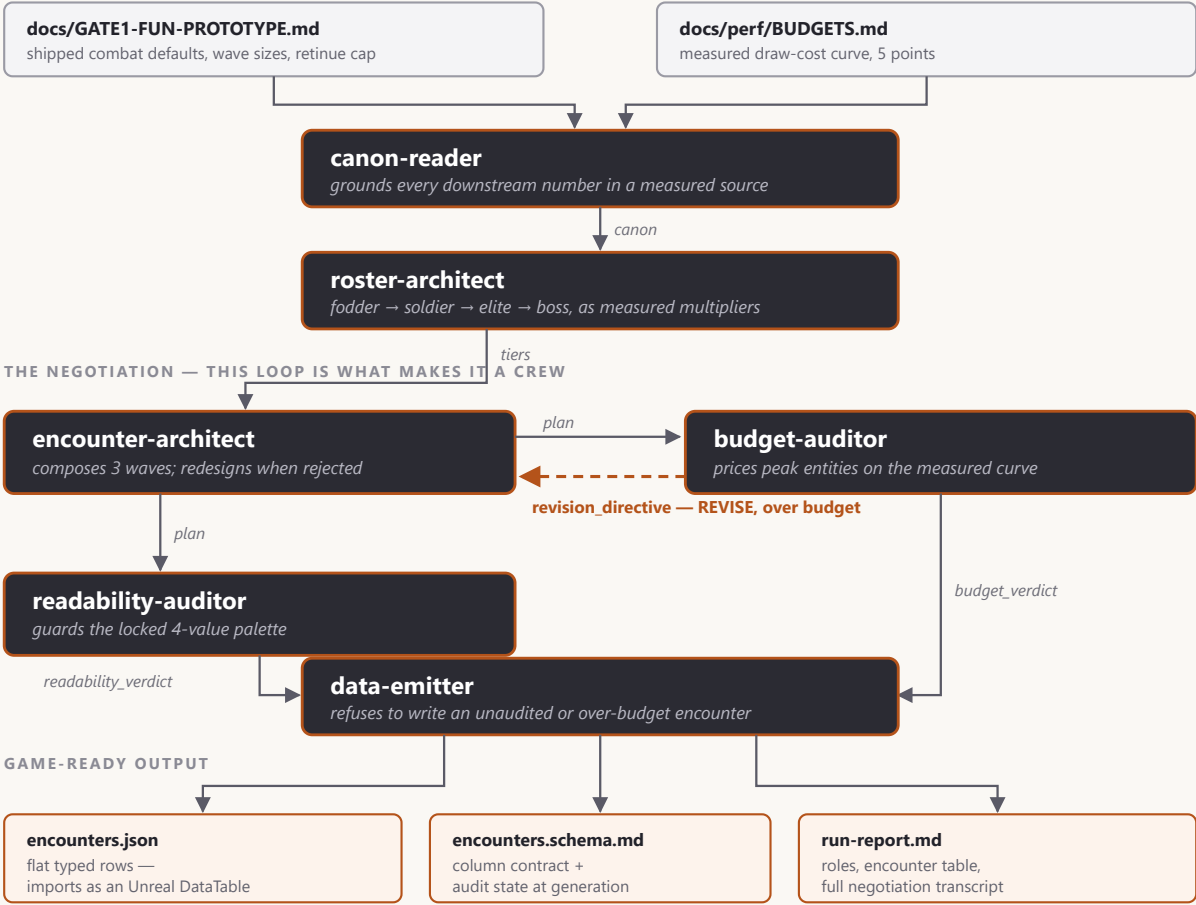
2 · What the crew produces

One run writes the wave composition for one floor of the vertical slice, priced against the real frame budget and checked against the game's locked 4-value palette:

ARTIFACT	WHAT IT IS
out/encounters.json	Flat DataTable-shaped rows, importable by the UE project
out/encounters.schema.md	Column contract — units, ranges, meaning
out/run-report.md	Verdicts, numbers, and the full inter-agent transcript

3 · Architecture

REPO CANON — REAL PROJECT FILES



Every agent declares the blackboard keys it may read and write. The Blackboard enforces those declarations at runtime, so removing any agent halts the run at the first consumer that needs its artifact — the graph above is executable, not drawn.

Data flow between the six agents. Source: `flowchart TD` in `crew/README.md`. Solid arrows carry named blackboard keys; the dashed return path is the budget auditor sending a plan back for redesign.

4 • The six agents

Every agent reads a named key and writes a named key. Nothing is passed implicitly — an agent cannot read a key it did not declare, which is what makes the ordering load-bearing rather than incidental.

AGENT	ROLE	READS	WRITES
canon-reader	Parses the project's shipped design and measured performance docs into one fact block, so no downstream agent invents a number.	—	canon
roster-architect	Derives the tier ladder (fodder → soldier → elite → boss) as multipliers over the measured combat baseline, and prices each tier.	canon	tiers
encounter-architect	Composes the three escalating waves and the spawn cadence — and redesigns when an auditor sends the plan back.	canon, tiers, revision_directive	plan
budget-auditor	Projects peak concurrent entities onto the measured cost curve. Returns PASS, or REVISE with a scale directive the architect must obey.	canon, plan	budget_verdict, revision_directive
readability-auditor	Checks each wave's simultaneous distinct silhouettes against the locked 4-value palette.	canon, plan	readability_verdict
data-emitter	Flattens the approved plan into DataTable rows, writes the column schema, and refuses to emit anything that has not passed both audits.	all of the above	artifacts

Why no agent is removable

canon-reader — every downstream number becomes invented. The crew would still run and produce a plausible-looking encounter with no grounding in the game.

roster-architect — the encounter architect has bodies but no tiers to compose from, and no per-tier cost to be audited against.

encounter-architect — nothing produces a **plan**; both auditors and the emitter have no input.

budget-auditor — the crew emits encounters it cannot draw. This is the agent that actually fired on the latest run, forcing a redesign.

readability-auditor — waves ship that are affordable but unparseable at speed, which the 4-value palette makes a real failure mode rather than a hypothetical one.

data-emitter — verdicts exist but nothing reaches the game. No artifact is produced at all.

5 · Evidence it runs

AGENTS 6	ROWS EMITTED 7	NEGOTIATION ROUNDS 1	BUDGET PASS	READABILITY CROWDED
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The crew does not merely run in sequence — the auditors have power over the designer. On the latest run the budget auditor rejected the first plan and issued a scale directive; the encounter architect recomposed the waves and the second plan passed. That round trip is the pipeline working as designed, and it is recorded in the transcript.

WAVE	BODIES	COMPOSITION	BUDGET PTS	SPAWN
1	250	fodder ×250	250	ring
2	450	fodder ×338, soldier ×112	674	ring
3	701	fodder ×420, soldier ×238, elite ×42, boss ×1	1698	arena_entrances

Budget verdict. Peak concurrent entities **821** (wave 3 plus the retinue cap of 120), projected at **11.12 ms** against the **16.6 ms** budget — **5.48 ms** of headroom, on the measured draw curve for a single client.

Readability verdict: CROWDED. Wave 3 puts four distinct enemy types on screen against two free palette values. The crew reports this rather than suppressing it — an honest warning the designer can act on is worth more than a silent pass.

canon-reader	-> wrote	canon
roster-architect	<- read	canon
roster-architect	-> wrote	tiers
encounter-architect	<- read	canon, tiers
encounter-architect	-> wrote	plan
budget-auditor	-> REVISE	(scale directive)
encounter-architect	-> wrote	plan (round 2)
budget-auditor	-> PASS	11.12ms / 16.6ms
readability-auditor	-> CROWDED	
data-emitter	-> 7 rows,	schema, report

6 · How the output reaches the game

encounters.json is emitted flat and typed, with one unique row key per entry, so it imports directly as an Unreal DataTable — the same shape the project's other design data uses. The schema file states each column's units and valid range, so a designer changing a number by hand knows what they are changing.

The connection runs both ways, and that is the point: the crew *reads* the game's real measured files and *writes* data the game consumes. It is not a general-purpose encounter generator pointed at a game — it is grounded in this project's own numbers, and its budget verdicts are only meaningful because those numbers were measured rather than assumed.